FARES IBRAHIM

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EDUCATION

Masters of Science in Computer Science, University of South Florida — Expected December 2026
GPA: 3.95
Bachelors of Science in Computer Engineering, University of South Florida — August 2020 - December 2024
GPA: 3.54
Relevant Coursework:Data Structures and Algorithms, Analysis of Algorithms, Embedded Systems, Operating Systems, Deep Learning, Machine Learning, Fair Machine Learning, Natural Language Processing

SKILLS

Languages: C/C++, Python, JavaScript, HTML/CSS, Java, SQL, Dart
Frameworks: React.js, Next.js, Flask, Django, Flutter, MySQL, PostgreSQL
Libraries: NumPy, Pandas, Sci-KitLearn,Matplotlib,Seaborn
Tools: GitHub, Kaggle, HuggingFace, Colab, SmolAgents, LangGraph, Jupyter Notebook

RESEARCH PUBLICATIONS

Anchorless Diversification for Parallel LLM Ideation — arXiv:2605.30150 — Under Review at EMNLP 2026 [Paper]

- First-author paper studying inference-time controls for diversifying LLM-generated candidate-idea pools
- compares independent generation, semantic direction stratification, and anchored baselines across creative task families.

RESEARCH EXPERIENCE

Adversarial Attacks on Large Language Models — Fall 2025
Research Project, University of South Florida — [Paper]

- Co-authored "Only The Elite: Your Model Doesn't Need Many Reasons to Be Evil" under supervision of Dr. Anshuman Chhabra
- Developed coverage-based demonstration selection method for adversarial in-context learning attacks on LLMs
- Achieved 80%+ attack success rates using minimal examples (k=2-3) on Vicuna 7B, Mistral 7B, and Llama 3 8B models
- Evaluated framework on AdvBench and Forbidden Prompts datasets, demonstrating critical vulnerabilities in model alignment

Fair Submodular Maximization in Streaming Settings — Spring 2025
Reproduction Study, University of South Florida — [Paper]

- Reproduced and extended "Fairness in Streaming Submodular Maximization: Algorithms and Hardness" under supervision of Dr. Anshuman Chhabra
- Implemented and evaluated Fair-Streaming, Fair-Sample-Streaming, and Fair-Greedy algorithms in C++
- Extended original work by validating algorithms on novel Amazon Sales dataset, demonstrating generalizability to commercial recommendation systems
- Analyzed fairness-utility tradeoffs across MovieLens, Census, and Banking datasets

PROJECTS

SoccerStats - Football AI Assistant (May 2025 - May 2025). Developed a comprehensive soccer statistics platform that tracks weekly games, analyzes player performance from Top 5 leagues including G/A ratios, and provides team information through an integrated AI chatbot. Tech stack: Python, Flask, HTML, CSS, SQLite, Open Source Football API, Fetch API AI's AS1-mini LLM. Implemented real-time data fetching for match schedules and player statistics, with an intelligent chatbot optimized for complex soccer-related queries. (Try it here)

ICalendar (May 2025 - May 2025). Developed a calendar app with an AI speech assistant , customer support style . A user can sign up and log their future events while also having access to a voice agent powered by Deepgram's speech-to-text API and receiving a response by another voice agent from ElevenLabs's text-to-speech API for easier task management , all done with an elegant user interface from Three js.Tech Stack : HTML , CSS, Three Js, Flask, SQLlite ,Deepgram API ,Elevenlabs API

Pain or No Pain Classifier (January 2025 - Febrauary 2025). Implemented a Deep Neural Network using Tensorflow/Keras to classify data obtained on 5 subject infants to assess their pain level. Tech stack: Python, TensorFlow, Keras, Numpy, Pandas, Matplotlib, Scikit-learn. Applied Adam Optimizer to smooth the loss function curve and evaluated the model's success using metrics like confusion matrix and accuracy (85% accuracy on test data). (Try it here)

Calories Burnt Predictor (March 2025 - March 2025) Used XGBoost Regressor model to predict a range of possibly burnt calories for a 1500 subject dataset from Kaggle. Tech stack: Python, XGBoost, Pandas, Numpy, Scikit-learn, Matplotlib, Seaborn. Evaluated model performance using metrics like Mean Absolute Error and Mean Squared Error Curve.(Try it here)

EXPERIENCE

Junior Developer Intern — May 2023 - August 2023
DecOps LLC — *Remote*

- Partnered with UX/UI designers to enhance user experience in React -based applications..
- Developed custom components and integrated third -party APIs using JavaScript.
- Created interactive animations with TailWind CSS for enhanced user engagement..

Student Research Assistant — January 2024 - Present
University of South Florida SHILED and CSSAI Labs — *Tampa,FL*

- Developed Python shell scripts for data extraction from Excel files using OpenPyXL.
- Preprocessed raw data from medical documents on Jupyter using Numpy and Pandas.
- Modeled relationships between data by training ML models to recognize patterns and assess long term predictions

EXTRA-CURRICULAR ACTIVITIES

- Treasurer for Muslims Student Association at University of South Florida.
  - Managing the organization's budget and financial records.
  - Ensuring compliance with university financial policies.
- CodePath Recurruing Mentor.
  - Mentored software engineering students by maintaining consistent communication and providing timely feedback within 48-hour response windows
  - Facilitated regular one-on-one sessions to address technical concerns, career questions, and skill development opportunities